



SFWRENG 20P3

Object-Oriented Programming

Dr. Lingyang Chu

Department of Computing and Software

McMaster University

Course Page & Class Schedule

- **Textbook:** *Java, Java, Java: Object-Oriented Problem Solving*, Written by: Ralph Morelli, Ralph Walde, Beryl Hoffman. ([link](#))
- **Class Homepage:** <https://avenue.cllmcmaster.ca/d2l/home/712234>
- **Lectures (In-person):**
 - 9:30AM – 10:20AM on Monday and Thursday.
 - 10:30AM – 11:20AM on Tuesday.
 - All lectures will be in person and will not be recorded.

Course Page & Class Schedule

Tutorials (In-person, not recorded):

T01: Monday 3:30PM – 4:20PM at ETB 230

TA: Reyhaneh Norouziani

T02: Thursday 8:30AM - 9:20AM at ETB 230

TA: Fiona(Fei) Xie

T03: Monday 12:30AM - 1:20PM at ETB 126

TA: Mehrdad Eshraghi Dehaghani

T04: Thursday 1:30PM - 2:20PM at ETB 230

TA: Tony Lin

Note: detailed information of the tutorials is on Avenue in the section of “General Course Information”.

Course Overview: Topics

- We are programming in Java.
 - We will use **Java** to introduce the concepts and principles of Object-Oriented Programming (OOP).
 - We will not use IDE. Instead, we will use command line and Maven to compile and run code.
 - *You will need a personal computer that can run Java.*
- Why Java not Python?
 - Both Java and Python are OOP programming languages, they both demonstrate the key principles of OOP.
 - There are many OOP languages. **Question: can you give me another example of OOP?**
 - The stricter syntax of Java better demonstrates the principles of OOP, which is better for beginners.
- Topics covered by this course include:
 - Classes, objects, interfaces, references, methods and fields, etc.
 - Types of inheritance, overriding, polymorphism, overloading, exception handling, generics, recursion, design by contract, file I/O and use of standard APIs, etc.
 - Some advanced OOP topics.

Course Work and Grading

Assignments, Quizzes, Projects and Exams

- Five regular assignments (3% each, 15% in total)
 - One project assignment (15%)
 - Mid-term exam (in-person invigilated, 30%)
 - Final exam (in-person invigilated, 40%)
 - The above grades sum up to 100%
 - *Bonus (Additional 5%, but the final grade will not exceed 100%.)*
- Note:** All works on assignments, programming projects, quizzes and exams is to be done individually. *No AI-assistant in any form is allowed.*

Need help and/or discussions?

- MS Teams Discussion Channel

Check your homework/exam scores:

- Avenue 2 Learn Course Website

How to Prepare and Submit Your Assignments

- All the assignments are posted on Avenue in Day 1.
- Each assignment has its own due time, which is shown in A2L.
- Each assignment may contain two types of questions: 'code' and 'text'

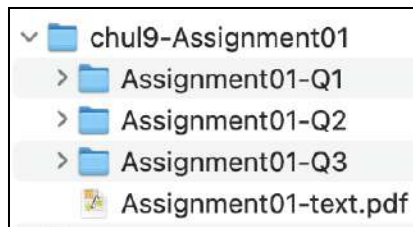
How to Prepare and Submit Your Assignments

- **How to answer a question of type 'code':** A question of type code requires to implement Java code and use Maven to produce executable jar file.
 - The answer to this question should be the root folder of a single Maven project, which can be directly compiled and packaged into a jar file that produces correct result.
 - The name of the root folder should be "Assignment?-Q?", where the question marks '?' should be replaced by the corresponding numbers of assignment and question.
 - Each question of type 'code' should correspond to one folder of a single Maven Project.
- **How to answer a question of type 'text':** A question of type text requires to provide your answer in plain text.
 - You need to include your answers to all the type 'text' questions in a single pdf file named "Assignment?-text", where the question mark '?' should be replaced by the corresponding number of the assignment.
 - In the pdf file, each answer should start with a header: "Assignment?-Q?", where the question marks '?' should be replaced by the corresponding numbers of assignment and question.

How to Prepare and Submit Your Assignments

Here is how to submit your answer to Avenue.

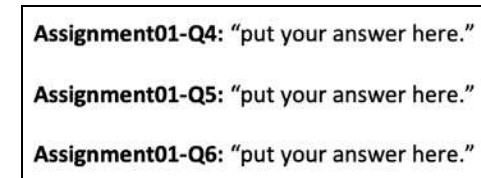
- You should only submit a single zip file named as “YourMacID-Assignment?.zip”, where ‘YourMacID’ is your MacID and ‘?’ is the number of the assignment.
- The zip file should contain a root folder named as “YourMacID-Assignment?”, which contains the folders of the Maven projects as the answers to the type-code questions and also the single pdf file as the answer to the type-text questions.



Example of folder structure



Example of zip file



Example of the content in the pdf file

IMPORTANT NOTES!

- Please follow exactly the above format to prepare and submit your answers. Violating the format is subject to a penalty of 20% of the marks of this assignment.
- Not all weeks have assignments. The due time of assignments may vary according to the posted information on Avenue 2 Learn (A2L).
- The grade of an assignment will be posted on A2L seven days after the due time. The answers will be explained in the tutorials next week.
- If you MSAF an assignment, the due time of the assignment will be extended, however, the extended deadline will not pass the time when the grades of the assignment are posted on A2L.
- An overdue submission is subject to a penalty of 20 marks per hour exceeding the submission deadline. The penalty is calculated by: $penalty = \text{floor}(\text{exceed hours}) * 20$ marks. For example, if you exceed 0.9 hours, the $penalty = \text{floor}(0.9) * 20 = 0$; if you exceed 2.1 hours, the $penalty = \text{floor}(2.1) * 20 = 40$.
- If you do not agree on any marks of an assignment or an exam, you can appeal within seven days after the marks are published by sending the instructor an email. After seven days, the marks will be finalized, and no change is available.

How to Get the Bonus Marks?

- By answering questions correctly in the following two Ways.
- **First**, I will randomly ask questions in lectures. **The first student who answers a question correctly will get 1 bonus mark.**
- **Second**, there will be students posting questions in MS Teams. **The first student who answers a meaningful posted question correctly will get 1 bonus mark.**
 - Dr. Chu preserves the rights to determine whether a posted question is meaningful or not.
 - **You should only send your email to claim a MS teams bonus mark when Dr. Chu @ you in Teams. And the same person will not be @ multiple times per day.**
 - For example: a question that is very similar to a previously answered questions will not be considered meaningful.

The bonus marks is in addition to the 100 marks of the course, but the final marks will not exceed 100.

For example:

- 1. If you get 85 marks in the course and get 5 bonus marks, then the final marks will be 90.***
- 2. If you get 98 marks in the course and get 5 bonus marks, then the final marks will be 100.***

How to claim a bonus mark?

- A bonus mark can be claimed by sending Dr. Lingyang Chu an email in **exactly the following format before 23:59 of the day when you gets the mark.**

Title of the email: 2OP3 bonus mark 2025

Content of the email:

Full Name: John Smith

MacID: john123

StudentID: 1234567

Description: a brief description about how you get this bonus mark.

example



☆ D dweller00.washes@icloud.com

2OP3 bonus mark 2025

To: Lingyang Chu

Full Name: John Smith

MacID: john123

StudentID: 1234567

Description: a brief description about how you get this bonus mark.

- The bonus mark will be **voided** if
 - The email format is not correct; **OR**
 - The claim email is not sent before 23:59 of the day when you gets the bonus mark.
 - The format is **case sensitive, thus capitalize the title and content of your email accordingly.**
- If you bonus mark is successfully claimed, you will receive a receipt email from Dr. Chu saying: “noted.”
 - Please keep the receipt emails until the end of the term.
 - Otherwise, you may not get the bonus marks added in your final grade.

Teaching Format

- **In Person Teaching**

- Lectures
- Tutorials
- Office hours

- **Avenue**

- Regular assignments and project assignments
- Grades of assignments and exams.
- Publish course related materials.

- **MS Teams**

- Announcements and notifications.
- Online discussions and Q&A.

Teaching Format: Avenue

Raw access to unorganized course related materials. (**NOT recommended to use**)

General information of the course such as time schedule, contact information etc.

Videos, slides, teaching progress will all be here. (**Recommend to check frequently.**)

Regulations and policies. (**Recommend to read carefully.**)

Assignments, quizzes and their due dates will be posted here. (**Recommend to check frequently.**)

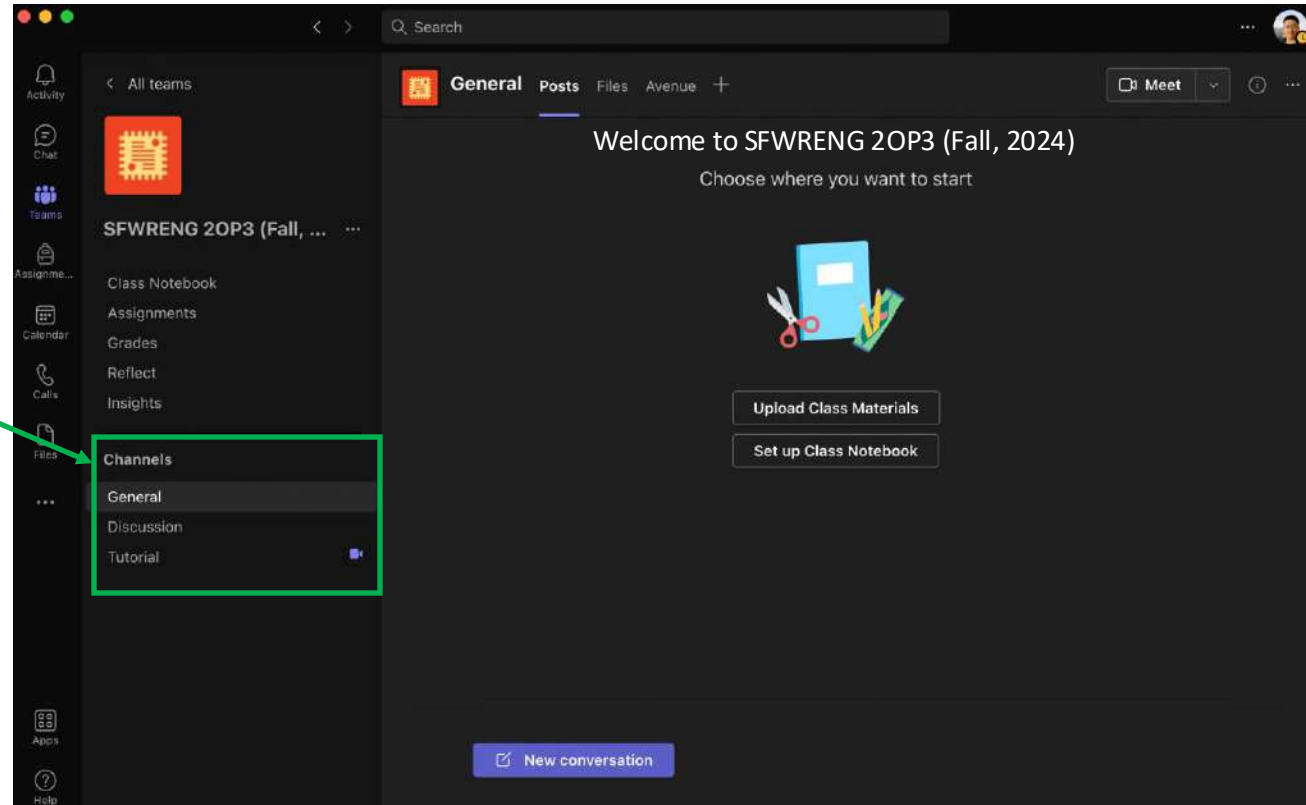
Link to the MS team of this course. **You can join the team by yourself using the link here.**

Missing the posting of assignments and quizzes will not be a valid reason for overdue submission.

Teaching Format: MS Teams

Usage of Channels

- **General:** important announcements will be posted here. Please check here frequently. **Missing the posting of an announcement will not be a valid excuse for anything.**
- **Discussion:** this is where you can ask questions to get answers from me, the TAs, and your classmates. **This is where you can get your bonus marks outside of classroom.**
- **Tutorial:** this is a temporary channel which is only used to deliver online tutorial under special conditions if there is any.



Academic Integrity

- Confirmed infringement to academic integrity will be reported to the office in charge and will cause a direct failure of this course.
- We will use professional software to detect code similarities and AI traces.
 - All the assignments detected as similar by the software will be equally considered as potential plagiarism. There is no difference between which is original, and which is not.
 - DO NOT use ChatGPT like AI software. If 10 people uses ChatGPT to do the same assignment, there is a good chance that many code pieces are similar. **WE CAN DETECT IT!**
- This is a course of coding, and you are getting a software engineering degree.
 - If one needs to cheat to succeed in this course, perhaps it is time to think whether it is a wise idea to proceed with the program.

Academic Integrity

- **It is your responsibility to understand what constitutes academic dishonesty.** (*Please refer to the Statement section on course website.*)
- Be aware of what is an Academic offense.
- Some typical examples:
 - Third party “tutorials” or “tutors”, or virtual tutors like ChatGPT
 - Online forums such as reddit and stack overflow.
 - Someone looking at your code or answers without your knowledge.
 - Hiring someone to do it for you.

Use Google to learn, do not use Google to copy and paste.

Tips to Success (in this course, and many others)

- The best way to learn coding is coding.
 - Practice coding as much as you can.
 - Do some interesting coding projects:
 - Use java to write a mini game, etc. **Group up a team with your best friends!**
- Classroom is where you learn the fundamentals, Internet is where you learn it all.
 - For example, use Github and Leetcode more to sharpen your coding skills.
 - Be creative in learning.
- Ask more questions, ask the right questions.
 - Asking a right question is 50% success to getting a right answer.
 - The more questions you ask, the more you learn.
 - Frequently asking right questions is an excellent sign of active thinking and learning.

Q & A